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Isothermal Elastohydrodynamic Lubrication of Point Contacts

Part IV—Starvation Results

Utilizing the theory and numerical procedure developed by the authors in an earlier publication the influence of lubricant starvation on minimum film thickness was investigated. This study of lubricant starvation was performed simply by moving the inlet boundary closer to the contact center. From the results it was found that for the range of conditions considered the value of dimensionless inlet distance at the boundary between fully flooded and starved conditions (m^) can be expressed simply as*

$$m^* = 1 + 3.06 \left[\left(\frac{R_x}{b} \right)^2 H_{c,F} \right]^{0.58}$$

or

$$m^* = 1 + 3.34 \left[\left(\frac{R_x}{b} \right)^2 H_{min,F} \right]^{0.56}$$

that is, for a dimensionless inlet distance (m) less than m^ , starvation occurs, and for $m \geq m^*$, a fully flooded condition exists. Furthermore, it has been possible to express the central and minimum film thickness for a starved condition as*

$$H_{c,S} = H_{c,F} \left(\frac{m - 1}{m^* - 1} \right)^{0.29}$$

$$H_{min,S} = H_{min,F} \left(\frac{m - 1}{m^* - 1} \right)^{0.25}$$

Contour plots of the pressure and film thickness in and around the contact are shown for the fully flooded and starved lubricant condition. From these contour plots it was observed that the pressure spike becomes suppressed and the film thickness decreases substantially as starvation increases.

Introduction

It was not until the late 1960s and early 1970s that the influence of lubricant starvation upon elastohydrodynamic behavior received serious consideration. Prior to this time it was assumed that the inlets were fully flooded. This assumption seemed to be entirely reasonable in view of the minute quantities of lubricant required to provide an adequate film. However, in due course it was recognized that some

machine elements suffered from lubricant starvation.

How partial filling of the inlet to an elastohydrodynamic conjunction influences pressure and film thickness can readily be explored theoretically by adopting different starting points for the inlet pressure boundary. Orcutt and Cheng [1]¹ appear to have been the first to proceed in this way for a specific case corresponding to a particular experimental situation. Their results showed that lubricant starvation lessened the film thickness. Wolveridge, et al. [2] used a Grubin approach [3] in an analysis of starved elastohydrodynamic lubricated line contacts. Wedeven, et al. [4] analyzed a starved condition in a ball on a plate geometry configuration and Castle and Dowson [5] pre-

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¹ Numbers in brackets designate References at end of paper.

sented a numerical solution for the starved line-contact elastohydrodynamic situation. In both references [2] and [4] the analysis yielded values of the proportional reduction in center-line film thickness from the fully flooded condition in terms of a dimensionless inlet boundary parameter.

Only in recent years have complete solutions of the isothermal elastohydrodynamic lubrication (EHL) of point contacts been presented. The analysis requires the simultaneous solution of the elasticity and Reynolds equation. The authors' approach to the theoretical solution is presented in two previous publications [6, 7].

The first of these publications (reference [6]) presents the elasticity model in which the contact zone is divided into equal rectangular areas, and a uniform pressure applied over each area is assumed. The second (reference [7]) gives the complete approach to solving the problem of elastohydrodynamic lubrication of point contacts.

The most important practical aspect of the EHL point contact theory (reference [7]) is the determination of the minimum film thickness within the contact. That is, the maintenance of a sufficient fluid film is extremely important to the operation of the machine elements in which these thin, continuous fluid films occur. In a recent paper by the authors (reference [8]) the fully flooded results obtained from the theory given in references [6, 7] were presented. In the results the influence of the ellipticity, dimensionless speed, load, and material parameters on minimum film thickness was investigated. Thirty-four different cases were used in obtaining the fully flooded minimum film thickness formula.

The present paper uses the basic theory developed in references [6, 7] in studying the effect of lubricant starvation on pressure and film thickness within the conjunction. The objective of this work is to provide a simple expression for the dimensionless inlet boundary distance. This inlet boundary distance defines whether a fully flooded or a starved condition exist in the contact. An additional objective was to develop simple expressions for the minimum and central film thickness under starvation condition. Fifteen cases in addition to three presented in reference [8] were used in obtaining the starvation results. To more fully understand what occurs in going from a fully flooded condition to a lubricant starvation condition contour plots of pressure and film thickness in and around the contact are shown. The theoretical findings are compared with previously reported results.

Boundary Between Fully Flooded and Starved Conditions

Fig. 1 shows the computing area in and around the Hertzian contact. In this figure the coordinate X is made dimensionless with respect to the semiminor axis (b) of the contact ellipse and the coordinate Y is made dimensionless with respect to the semimajor axis (a) of the contact ellipse. The ellipticity parameter (k) is defined as the semi-

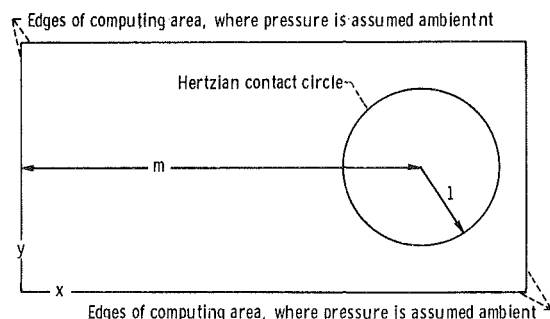


Fig. 1 Computing area in and around the Hertzian contact circle

major axis divided by the semiminor axis of the contact ellipse ($k = a/b$). Because of the dimensionless form of the coordinates X and Y , the Hertzian contact ellipse becomes a Hertzian circle regardless of the ellipticity parameter. This Hertzian contact circle is shown in Fig. 1 with a radius of unity. The edges of the computing area, where the pressure is assumed to be ambient, are also denoted. In this figure the dimensionless inlet distance (m), which is equal to the dimensionless distance from the center of contact to the inlet edge of the computing area, is shown.

Lubricant starvation can be studied simply by reducing the dimensionless inlet distance (m). A fully flooded condition is said to exist when the dimensionless inlet distance (m) ceases to influence in any significant way the minimum film thickness. When starting from a fully flooded condition and decreasing (m), the value at which the minimum film thickness first starts to change is called the fully flooded-starved boundary and is denoted by m^* . Therefore, lubricant starvation was studied by using the basic elastohydrodynamic lubrication point contact theory developed in reference [7] and observing the effect of reducing the dimensionless inlet distance (m).

Table 1 shows the effect of changing the dimensionless inlet distance upon the dimensionless minimum film thickness for three groups of dimensionless load and speed parameters. For all the results presented in this paper the material parameter (G) is fixed at 4522 and the ellipticity parameter is fixed at 6. In this table it is seen that as the dimensionless inlet distance (m) decreases the dimensionless minimum film thickness ($H_{\min}$) decreases.

Table 2 shows how the three groups of dimensionless speed and load parameters affect the location of the dimensionless inlet boundary distance (m^*). Also given in this table are the corresponding values of the dimensionless central and minimum film thickness for the fully flooded condition as obtained by interpolation of the numerical values. The value of the dimensionless inlet boundary (m^*) shown in Table

Nomenclature

A = constant defined in equation (2)

a = semimajor axis of contact ellipse

B = constant defined in equation (2)

b = semiminor axis of contact ellipse

E = modulus of elasticity

$$E' = \frac{2}{\left(\frac{1 - \nu_A^2}{E_A} + \frac{1 - \nu_B^2}{E_B}\right)}$$

F = normal applied load

G = dimensionless material parameter, $E'/p_{iv,as}$

$H_{c,F}$ = dimensionless film thickness at the center of the contact for a fully flooded conjunction, h_c/R_x

$H_{\min}$ = dimensionless minimum film thickness, $h_{\min}/R_x$

$H_{\min,F}$ = dimensionless minimum film

thickness for a fully flooded conjunction, $h_{\min,F}/R_x$

$\tilde{H}_{\min,S}$ = dimensionless minimum film thickness for a starved conjunction as obtained from the least square fit of the data, $\tilde{h}_{\min,S}R_x$

h = film thickness

k = ellipticity parameter, a/b

m = dimensionless inlet distance

m^* = value of dimensionless inlet distance at the boundary between fully flooded and starved conditions

m_w = dimensionless inlet distance boundary as obtained from Wedeven, et al. [4]

P = dimensionless pressure, p/E'

p = pressure

$p_{iv,as}$ = asymptotic isoviscous pressure

R = effective radius of curvature

r = radius of curvature

U = dimensionless speed parameter, $u\eta_0/E'R_x$

u = surface velocity in rolling direction

W = dimensionless load parameter, $F/E'R_x^2$

$x, X, \bar{X}$
 $y, Y, \bar{Y}$ = coordinate systems defined in paper

α = pressure viscosity coefficient

η_0 = atmospheric viscosity

ν = Poisson's ratio

Subscripts

A = refers to solid A

B = refers to solid B

x, y = refers to coordinate system

Table 1 Effect of starvation on minimum film thickness for three groups of the dimensionless speed and load parameters

Dimensionless inlet distance, m	Group		
	1	2	3
	Dimensionless load parameter, W		
	0.3686×10^{-6}	0.7371×10^{-6}	0.7371×10^{-6}
	Dimensionless speed parameter, U		
	0.1683×10^{-11}	1.683×10^{-11}	5.050×10^{-11}
Minimum film thickness, $H_{\min}$			
6	-----	29.75×10^{-6}	61.32×10^{-6}
4	6.317×10^{-6}	29.27	57.50
3	6.261	27.84	51.70
2.5	-----	26.38	46.89
2	5.997	23.46	39.91
1.75	-----	21.02	34.61
1.5	5.236	-----	27.90
1.25	3.945	-----	-----

Table 2 Effect of dimensionless speed and load parameter on dimensionless inlet distance at fully flooded-starved boundary

Group	Dimensionless speed parameter, U	Dimensionless load parameter, W	Dimensionless radius parameter, R_x/b	Fully flooded central film thickness, $H_{c,F}$	Fully flooded minimum film thickness, $H_{\min,F}$	Dimensionless inlet distance at fully flooded-starved boundary, m^*
1	0.1683×10^{-11}	0.3686×10^{-6}	205.9	7.480×10^{-6}	6.221×10^{-6}	2.62
2	1.683	.7371	163.5	33.55	29.20	3.71
3	5.050	.7371	163.5	70.67	60.92	5.97

2 was obtained by using the data from Table 1 when the following equation was satisfied:

$$\frac{H_{\min,F} - (H_{\min})_{m=m^*}}{H_{\min,F}} = 0.03 \quad (1)$$

The value of 0.03 is used in equation (1) since it was ascertained that the data in Table 1 could only be obtained to an accuracy of 3 percent.

The general form of the equation that describes how the dimensionless inlet distance at the fully flooded-starved boundary (m^*) varies is given as

$$m^* - 1 = A \left[\left(\frac{R_x}{b} \right)^2 H_{c,F} \right]^B \quad (2)$$

The right side of equation (2) is similar to the forms of the equation given by Wolveridge, et al. [2] and Wedeven, et al. [4]. Using the data obtained from Table 1, the following can be written:

$$m^* = 1 + 3.06 \left[\left(\frac{R_x}{b} \right)^2 H_{c,F} \right]^{0.58} \quad (3)$$

A fully flooded condition exists when $m \geq m^*$, and a starved condition exists when $m < m^*$.

If in equation (2) the dimensionless minimum film thickness is used instead of the central film thickness, the following is obtained:

$$m^* = 1 + 3.34 \left[\left(\frac{R_x}{b} \right)^2 H_{\min,F} \right]^{0.56} \quad (4)$$

From Wedeven, et al. [4], using the symbols of this paper, the dimensionless inlet distance at the fully flooded-starved boundary can be written as

$$m_w = 1 + 3.52 \left[\left(\frac{R_x}{b} \right)^2 H_{c,F} \right]^{2/3} \quad (5)$$

Comparing equation (3) with equation (5) indicates close agreement with Wedeven, et al. [4]. The latter, however, predicts a slightly higher value for the location of the fully flooded-starved boundary than predicted from the present results.

Film Thickness Formulas

Having clearly established the limiting location of the inlet boundary for the fully flooded conditions (equations (3) and (4)), an equation defining the dimensionless central film thickness for lubricant starvation conditions will be developed. The relationship between the dimensionless central film thickness in starved and fully flooded conditions can be expressed in general form as

$$\frac{H_{c,S}}{H_{c,F}} = C \left(\frac{m-1}{m^*-1} \right)^D \quad (6)$$

Table 3 shows how the ratio of the dimensionless inlet distance parameter to the fully flooded-starved boundary $[(m-1)/(m^*-1)]$ affects the ratio of central film thickness in the starved and fully flooded conditions ($H_{c,S}/H_{c,F}$). A least-square power curve fit to the 16 pairs of data points

$$\left[\left(\frac{H_{c,S}}{H_{c,F}} \right)_i, \left(\frac{m-1}{m^*-1} \right)_i \right], \quad i = 1, 2, \dots, 16$$

was used in obtaining the dimensionless central film thickness for a starved condition given as

$$H_{c,S} = H_{c,F} \left(\frac{m-1}{m^*-1} \right)^{0.29} \quad (7)$$

By using a similar approach while making use of the data in Table 3, the dimensionless minimum film thickness for a starved condition can be written as

$$H_{\min,S} = H_{\min,F} \left(\frac{m-1}{m^*-1} \right)^{0.25} \quad (8)$$

Therefore, whenever $m < m^*$, where m^* is defined by either equation (3) or (4), a lubricant starvation condition exists. When this is true, the dimensionless central film thickness is expressed by equation (7) and the dimensionless minimum film thickness is expressed in equation (8). If $m \geq m^*$, where m^* is defined by either equation (3) or (4), a fully flooded condition exists. The dimensionless central and minimum film thickness for a fully flooded condition ($H_{c,F}$ and $H_{\min,F}$) were developed in reference [8] and can be written as

$$H_{c,F} = 2.69 U^{0.67} G^{0.53} W^{-0.067} (1 - 0.61 e^{-0.73k}) \quad (9)$$

$$H_{\min,F} = 3.63 U^{0.68} G^{0.49} W^{-0.073} (1 - e^{-0.68k}) \quad (10)$$

where

$$U = \frac{\eta_0 \mu}{E' R_x} \text{ dimensionless speed parameter} \quad (11)$$

$$W = \frac{F}{E' R_x^2} \text{ dimensionless load parameter} \quad (12)$$

$$G = \frac{E'}{P_{iv,as}} \text{ dimensionless material parameter} \quad (13)$$

$$k = \frac{a}{b} \text{ dimensionless ellipticity parameter} \quad (14)$$

Equations (13) and (14) can be expressed in a more usable form as

$$G = E' \alpha \quad (15)$$

$$k = 1.03 \left(\frac{R_y}{R_x} \right)^{0.64} \quad (16)$$

In equation (16) the ellipticity parameter is expressed strictly in terms of the radii of curvature, thereby eliminating the common practice of evaluating the elliptical integrals of the first and second kind.

The ratio of the dimensionless inlet distance to the fully flooded-starved boundary as obtained from Wedeven, et al. (reference [4]), expressed as $\{(m-1)/(m_w-1)\}$, is also given in Table 3. By comparing these results with the results obtained from the present work $[(m-1)/(m^*-1)]$, it can be seen that for group 1 the agreement is excellent. However, the agreement in groups 2 and 3 is not as good. A possible explanation for this difference can be that in the Wedeven, et al. (reference [4]) analysis, an approximate expression for the Hertzian, deformation is used. They indicate that their equation

Table 3 Effect of dimensionless inlet distance on dimensionless central film thickness ratios

Group	Dimensionless inlet distance, m	Ratio of central film thicknesses for starved and flooded conditions, $H_{c,s}/H_{c,f}$	Ratio of minimum film thicknesses for starved and flooded conditions, $H_{min,s}/H_{min,f}$	Inlet boundary parameter, $(m-1)/(m^*-1)$	Inlet boundary parameter of Wedeven, et al. (1971), $(m-1)/(m_w-1)$
1	2.62	1	1	1	0.9895
	2	.9430	.9640	.6173	.6108
	1.5	.7697	.8417	.3086	.3054
	1.25	.5689	.6341	.1543	.1527
2	3.71	1	1	1	.8281
	3	.9574	.9534	.7380	.6111
	2.5	.8870	.9034	.5535	.4584
	2	.7705	.8034	.3690	.3056
	1.75	.7151	.7199	.2768	.2292
3	5.57	1	1	1	.8498
	4	.9348	.9439	.6565	.5579
	3	.8330	.8487	.4376	.3719
	2.5	.7440	.7697	.3282	.2789
	2	.6223	.6551	.2188	.1860
	1.75	.5309	.5681	.1641	.1395
	1.5	.4155	.4580	.1094	.0930

(equation (5)) is only valid for small m^* or more specifically $m^* < 3$. In group 2, $m^* = 3.71$ and in group 3, $m^* = 5.57$. Since no such assumption is required in deriving equations (3) and (4), they would seem to be more appealing.

Fig. 2 shows the influence of inlet boundary parameter upon central film thickness for the Wedeven, et al. (reference [4]) results and those obtained from the present work. From this figure it is observed that the Wedeven, et al. (reference [4]) results give slightly higher values of the central film thickness ratio of starved to fully flooded conditions than those obtained from the present results.

Contour Plot of Results

To explain more fully what happens in going from a fully flooded to a lubricant starvation condition, Figs. 3–8 are presented. In Figs. 3(a), (b), and (c) contour plots of dimensionless pressure ($P = p/E'$) are given for group 1 of Table 2 and for dimensionless inlet distances (m) of 4, 2, and 1.25, respectively. In these figures and the remainder of the contour plots to be presented the symbol + indicates the center of the Hertzian contact. It should be remembered that because of the way the coordinates are made dimensionless, the actual contact ellipse becomes a Hertzian circle regardless of the ellipticity parameter (k). The Hertzian contact circle is shown in each figure by asterisks. At the top of each figure the contour label and its corresponding value are given. The inlet region is to the left and the exit region is to the right.

In Figs. 3(a), (b), and (c) the contour values were kept constant. In Figs. 3(a) and (b) the pressure spikes are shown with the (B) and (C) contour labels, respectively, and in Fig. 3(c) no pressure spike occurs. This implies that as the dimensionless inlet distance (m) decreases (or as starvation occurs) the pressure spike is suppressed. In Fig. 3(a) a fully flooded conditions exists. In Fig. 3(b), with $m = 2$, a starved condition exists and in Fig. 3(c) with $m = 1.25$ the starvation is even more severe. That is, once starvation occurs the severity of the situation increases as (m) is decreased. Note in Fig. 3(c), the most severe starvation case, that the contour labeled (H) does not extend as far to the left as it did for the fully flooded pressure results shown in Fig. 3(a).

In Figs. 4(a), (b), and (c) contour plots of dimensionless film thickness ($H = h/R_x$) are given for group 1 of Table 2 and values of the inlet distance (m) of 4, 2, and 1.25, respectively. These film thickness results shown in Fig. 4(a), (b), and (c) correspond to the pressure results shown in Figs. 3(a), (b), and (c). The central portion of the film thickness contours has become more parallel as starvation has increased (Figs. 4(a)–(c)), with the minimum film area decreasing. Note also that the labels of the film thickness contour values for the starved condition (Fig. 4(c)) are much lower than those of the film thickness contour values of the fully flooded conditions (Fig. 4(a)).

In Figs. 5(a), (b), and (c) contour plots of dimensionless pressure ($P = p/E'$) are given for group 3 of Table 2 and dimensionless inlet distances of 6, 2.5, and 1.5, respectively. The contour values in Figs.

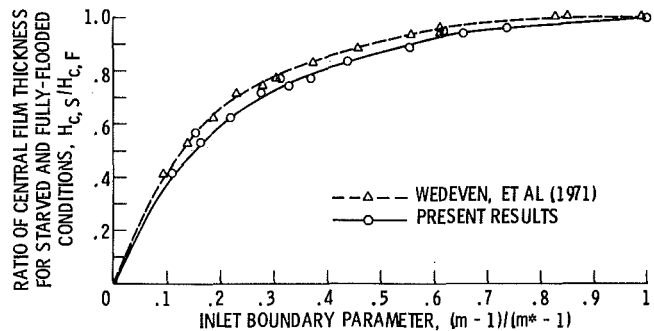


Fig. 2 Influence of inlet boundary parameters upon central film thickness ratio

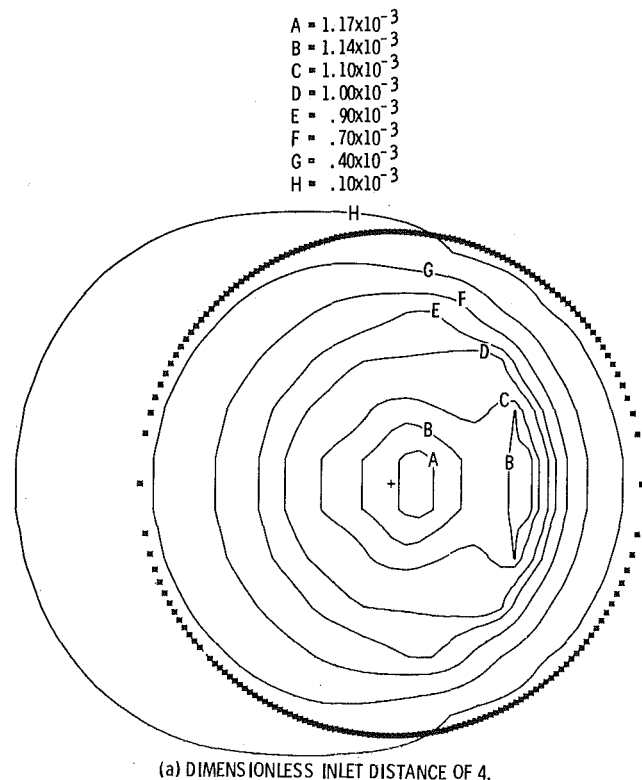


Fig. 3 Contour plot of dimensionless pressure for dimensionless inlet distances of 4, 2, and 1.25 and for group 1 of Table 2

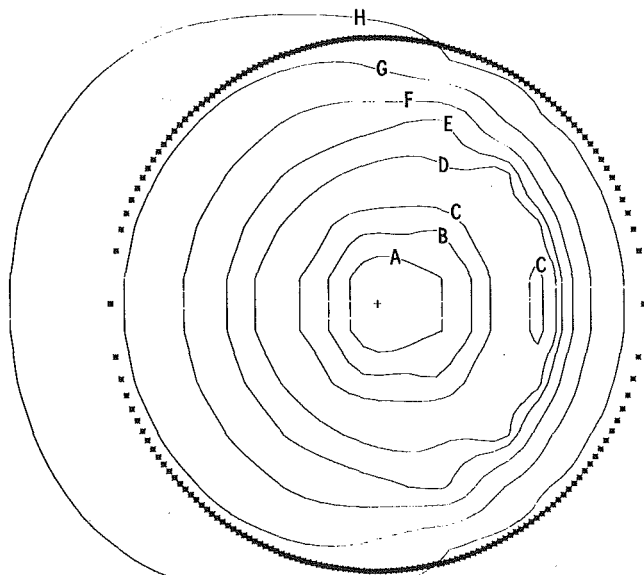
5(a), (b), and (c) are the same in each case. In Fig. 5(a) a fully flooded condition exists while in Fig. 5(c) a severely starved condition exists. The following observations can be made about Figs. 5(a), (b), and (c):

1 The distance from the center of the contact to the upstream location of the largest contour (labeled H) decreases as the severity of lubricant starvation increases.

2 Contours A, B, and C are absent in the severely starved condition (Fig. 5(c)).

Contour plots of the dimensionless film thickness ($H = h/R_x$) are shown for group 3 of Table 2 and for dimensionless inlet distances of 6, 2.5, and 1.5, respectively, in Figs. 6(a), (b), and (c). These film thickness results correspond to the pressure results shown in Figs. 5(a), (b), and (c). In Fig. 6(a) (the fully flooded condition) the minimum film area occurs to the sides of the conjunction in two areas which are midway between the center of the contact and the Hertzian circle. In Fig. 6(c) (the severely starved condition) the central portion has roughly parallel contours in the direction of motion with one minimum area directly behind the axial center of the contact and near

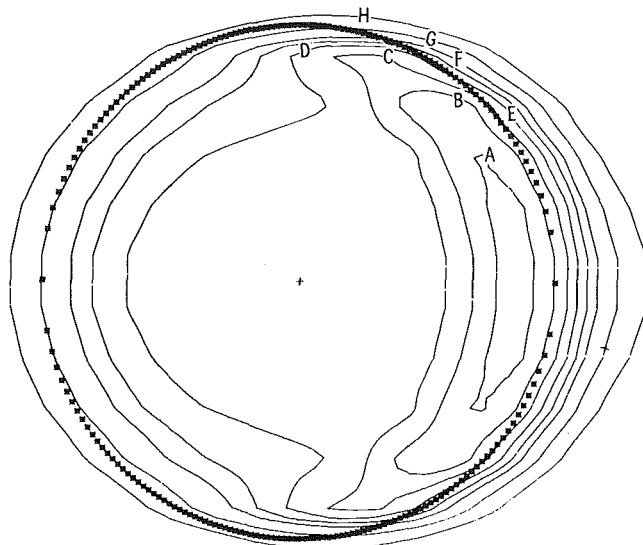
A = 1.17×10^{-3}
 B = 1.14×10^{-3}
 C = 1.10×10^{-3}
 D = 1.00×10^{-3}
 E = $.90 \times 10^{-3}$
 F = $.70 \times 10^{-3}$
 G = $.40 \times 10^{-3}$
 H = $.10 \times 10^{-3}$



(b) DIMENSIONLESS INLET DISTANCE OF 2.

Fig. 3 (continued)

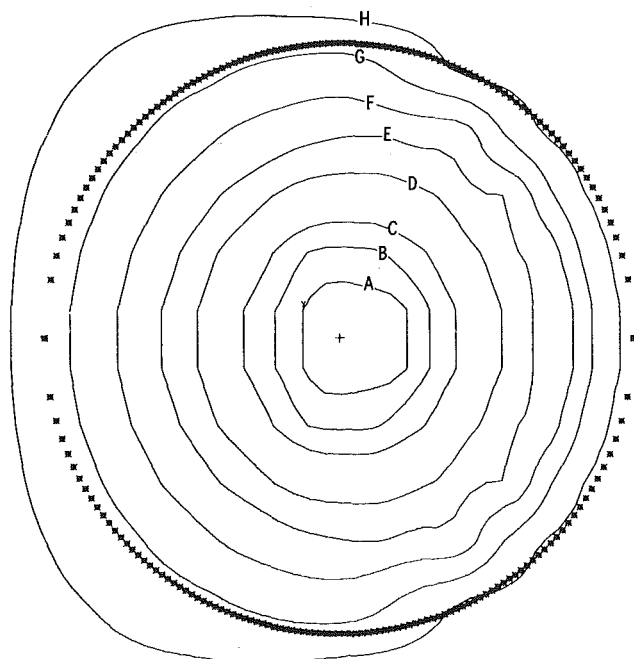
A = 6.5×10^{-6}
 B = 7.0×10^{-6}
 C = 7.5×10^{-6}
 D = 8.0×10^{-6}
 E = 8.5×10^{-6}
 F = 8.0×10^{-6}
 G = 10.0×10^{-6}
 H = 11.5×10^{-6}



(a) DIMENSIONLESS INLET DISTANCE OF 4.

Fig. 4 Contour plot of dimensionless film thickness for dimensionless inlet distances 4, 2, and 1.25 and for group 1 of Table 2

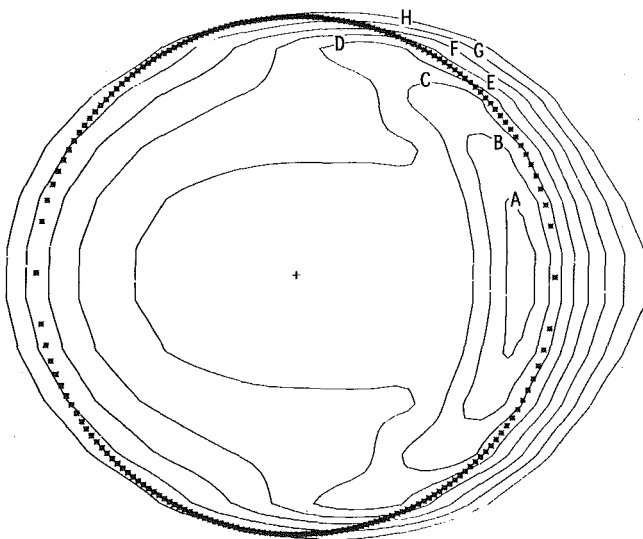
A = 1.17×10^{-3}
 B = 1.14×10^{-3}
 C = 1.10×10^{-3}
 D = 1.00×10^{-3}
 E = $.90 \times 10^{-3}$
 F = $.70 \times 10^{-3}$
 G = $.40 \times 10^{-3}$
 H = $.10 \times 10^{-3}$



(c) DIMENSIONLESS INLET DISTANCE OF 1.25.

Fig. 3 (concluded)

A = 6.1×10^{-6}
 B = 6.4×10^{-6}
 C = 6.8×10^{-6}
 D = 7.3×10^{-6}
 E = 7.9×10^{-6}
 F = 8.6×10^{-6}
 G = 10.3×10^{-6}



(b) DIMENSIONLESS INLET DISTANCE OF 2.

Fig. 4 (continued)

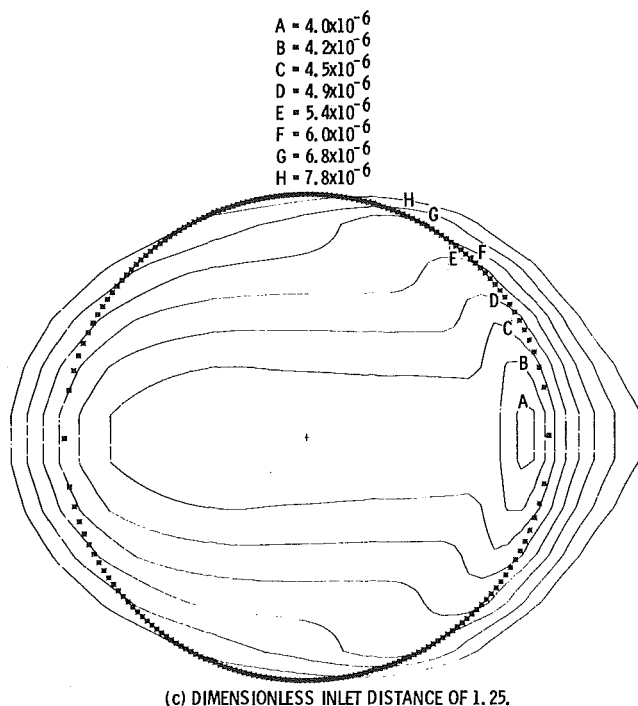


Fig. 4 (concluded)

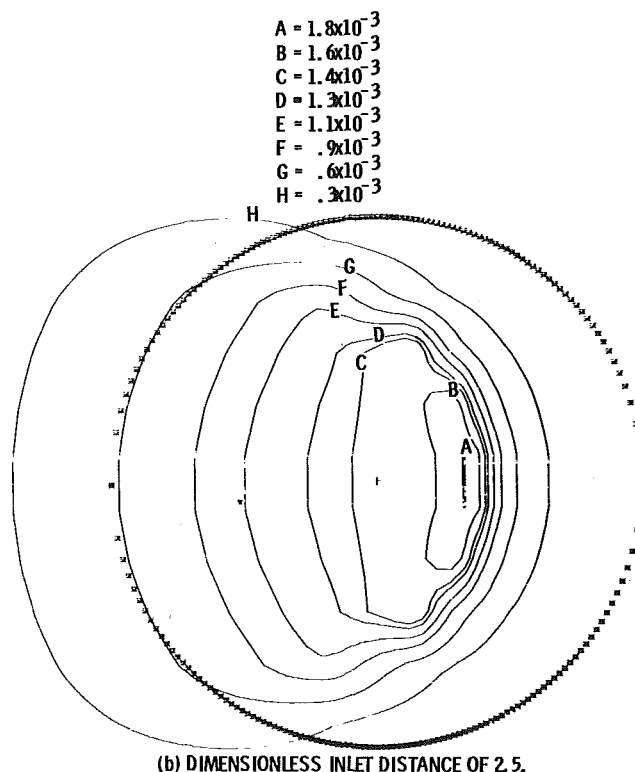


Fig. 5 (continued)

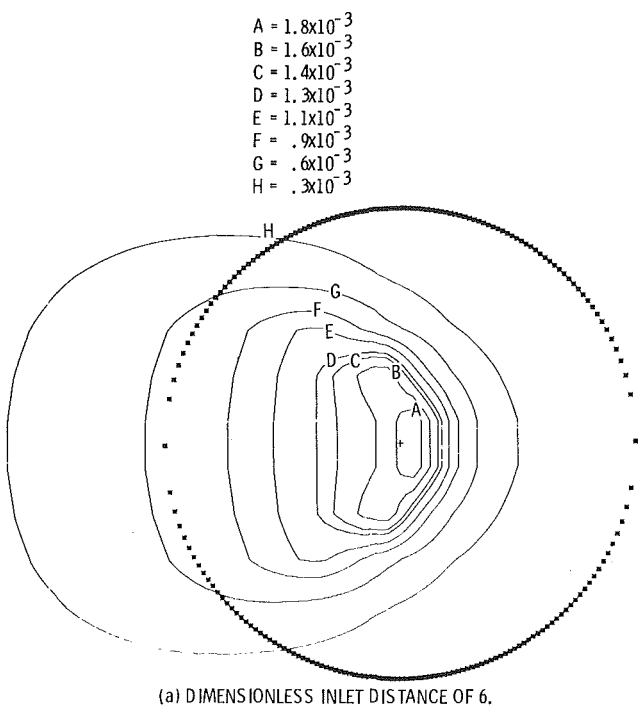


Fig. 5 Contour plot of dimensionless pressure for dimensionless inlet distances of 6, 2.5, and 1.5 and for group 3 of Table 2

the Hertzian circle. Ranger, et al. (reference [9]) also found a similar contour plot in their 1975 publication. Note the similarity between the film thickness contours of Fig. 4(c) (group 1 of Table 2) and that of Fig. 6(c) (group 3 of Table 2). From the labels of the contours in Figs. 6(a), (b), and (c) we see that the film thickness for the starved condition is much lower than the film thickness in the fully flooded condition.

The dimensionless pressure ($P = p/E'$) on the X axis is shown for three values of dimensionless inlet distance (m) and for groups 1 and

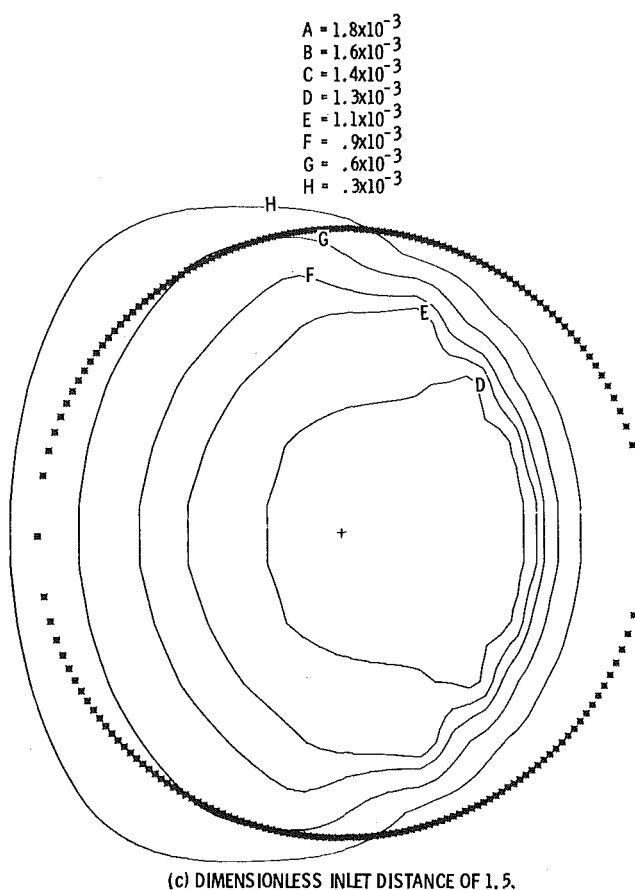


Fig. 5 (concluded)

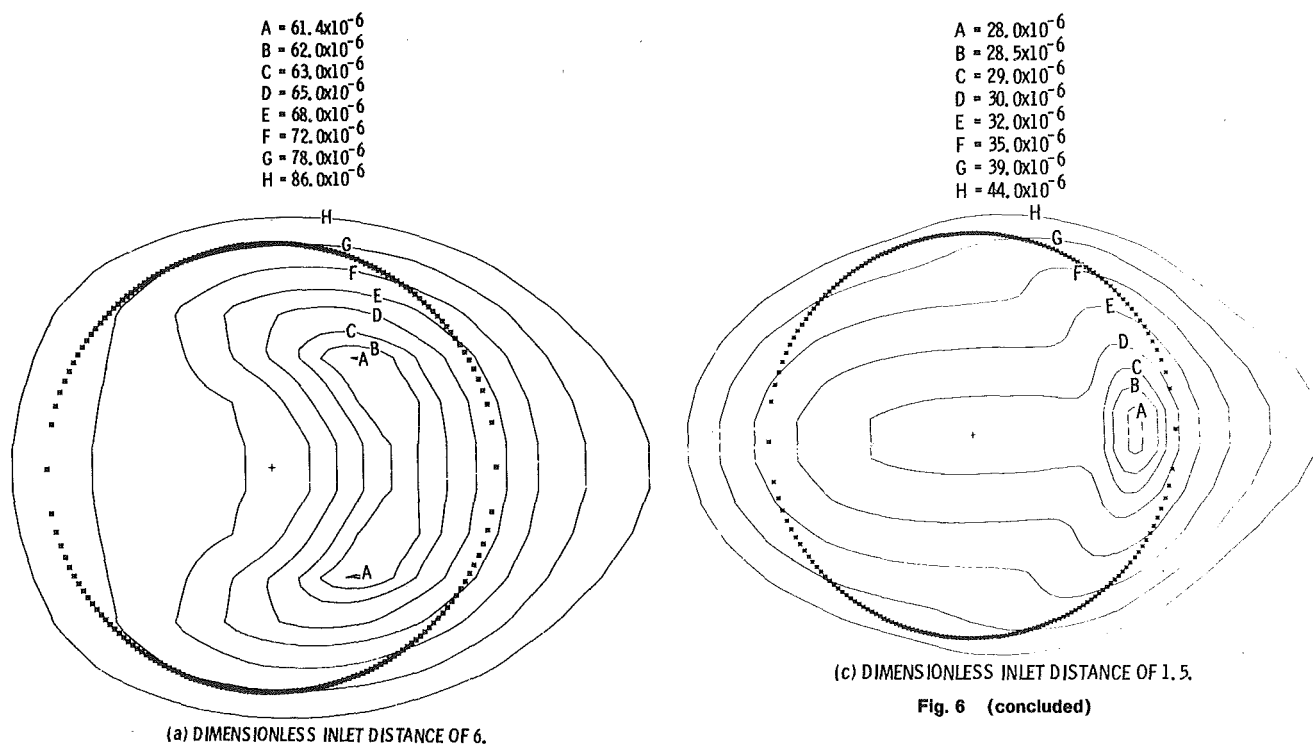


Fig. 6 Contour plot of dimensionless film thickness for dimensionless inlet distances of 6, 2.5, and 1.5 and for group 3 of Table 2

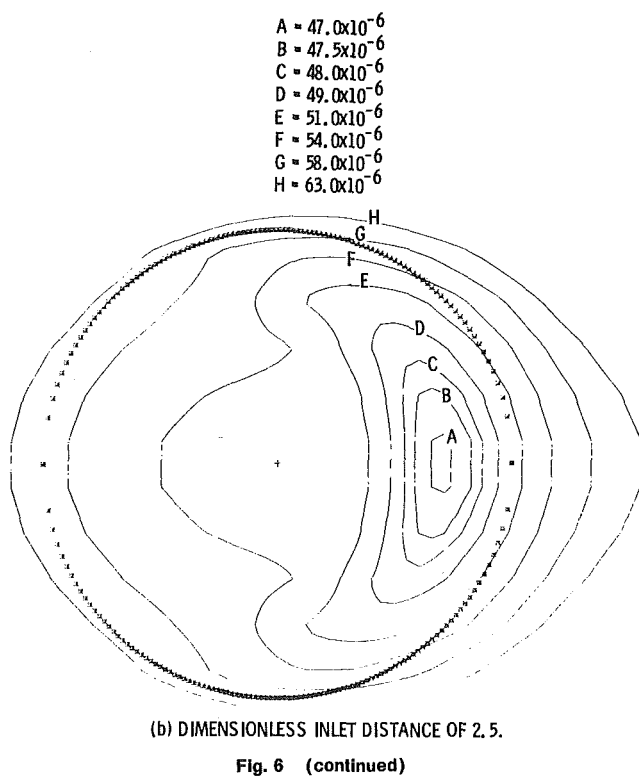


Fig. 6 (continued)

3 of Table 2, respectively, in Figs. 7(a) and (b). The value of Y is held constant near to the axis of symmetry of the conjunction. In these figures as a conjunction becomes starved (as m is decreased) the pressure spike diminishes.

Figs. 8(a) and (b) show the dimensionless film thickness ($H = h/R_x$) on the X axis for three values of the dimensionless inlet distance and for group 1 and 3 of Table 2, respectively. The value of Y is held fixed

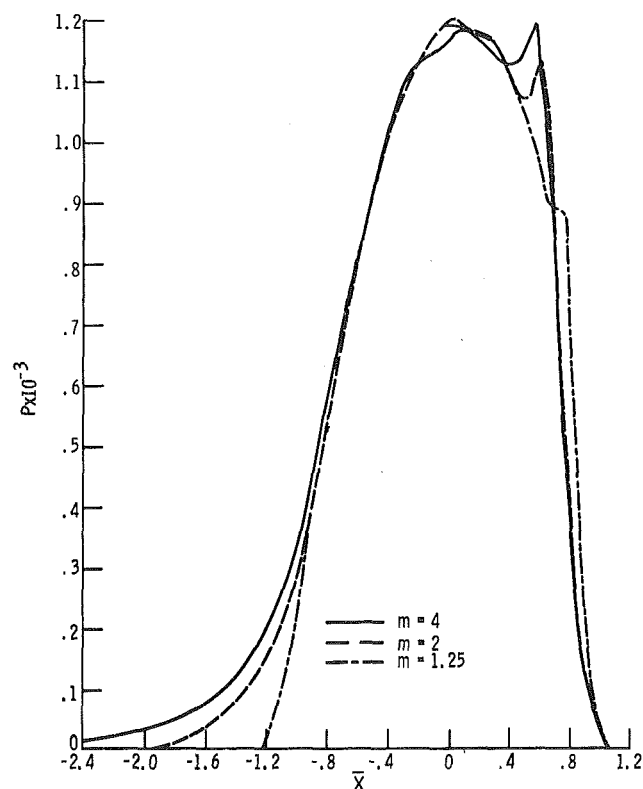


Fig. 7(a) Dimensionless pressure ($P = p/E'$) on the X axis for the values of dimensionless inlet distance (m). The value of Y is held fixed near the axis of symmetry of the contact. The dimensionless speed parameter (U) and load parameter (W) are held fixed as shown in group 1 of Table 2.

close to the axis of symmetry of the contact. In these figures, particularly Fig. 8(b), the central region becomes flatter as starvation occurs. Also, in going from a fully flooded condition to a starved condition the film thickness decreases substantially.

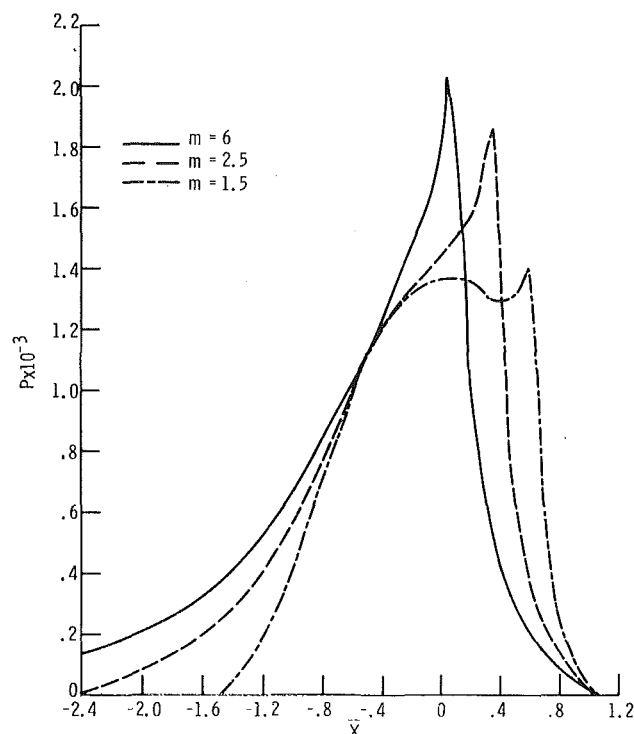


Fig. 7(b) Dimensionless pressure ($P = p/E'$) on the $\bar{X}$ axis for the values of dimensionless inlet distance (m). The value of $\bar{Y}$ is held fixed near the axes of symmetry of the contact. The dimensionless speed and local parameters are held fixed as shown in group 3 of Table 2.

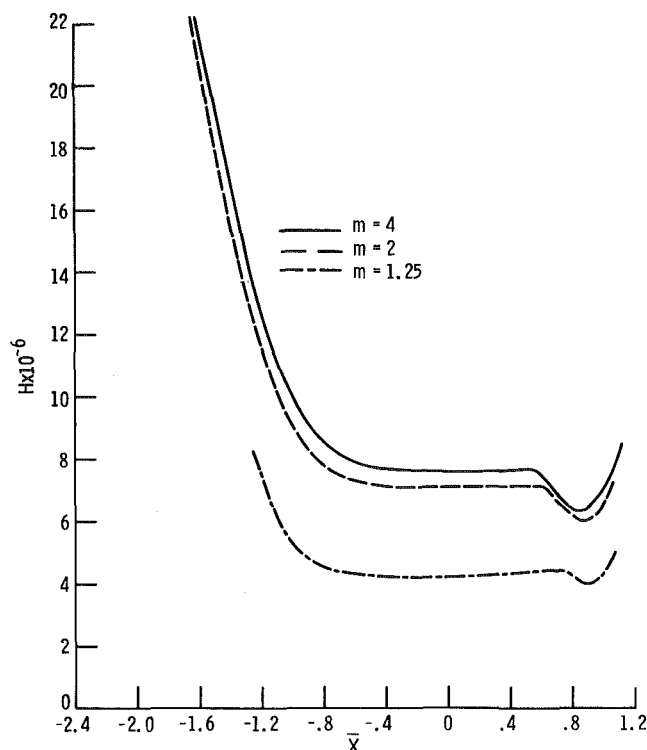


Fig. 8(a) Dimensionless film thickness ($H = h/R_x$) on the $\bar{X}$ axis for three values of dimensionless inlet distance (m). The value of $\bar{Y}$ is held fixed near the axis of symmetry of the contact. The dimensionless speed and load parameters are held fixed as shown in group 1 of Table 2.

Conclusion

By using the theory and numerical procedure outlined and developed by the authors in earlier publications (references [6, 7]) the influence of lubricant starvation upon minimum film thickness in starved elliptical elastohydrodynamic conjunctions has been investigated. This study of lubricant starvation was performed simply by moving the inlet boundary closer to the center of the conjunction. From the results it was found that the location of the dimensionless inlet boundary (m^*) between fully flooded and starved conditions could be expressed simply as

$$m^* = 1 + 3.06 \left[\left(\frac{R_x}{b} \right)^2 H_{c,F} \right]^{0.58}$$

or

$$m^* = 1 + 3.34 \left[\left(\frac{R_x}{b} \right)^2 H_{\min,F} \right]^{0.56}$$

That is, for a dimensionless inlet distance (m) less than m^* , starvation occurs and for $m \geq m^*$, a fully flooded condition exists. Furthermore, it has been possible to express the central and minimum film thickness for a starved condition as

$$H_{c,S} = H_{c,F} \left(\frac{m-1}{m^*-1} \right)^{0.29}$$

$$H_{\min,S} = H_{\min,F} \left(\frac{m-1}{m^*-1} \right)^{0.25}$$

where

$H_{c,F}$ = fully flooded dimensionless central film thickness

$H_{\min,F}$ = fully flooded dimensionless minimum film thickness

m = dimensionless inlet distance

m^* = dimensionless inlet distance at the fully flooded-starved boundary

Contour plots of the pressure and film thickness in and around the contact have been presented for both the fully flooded and starved

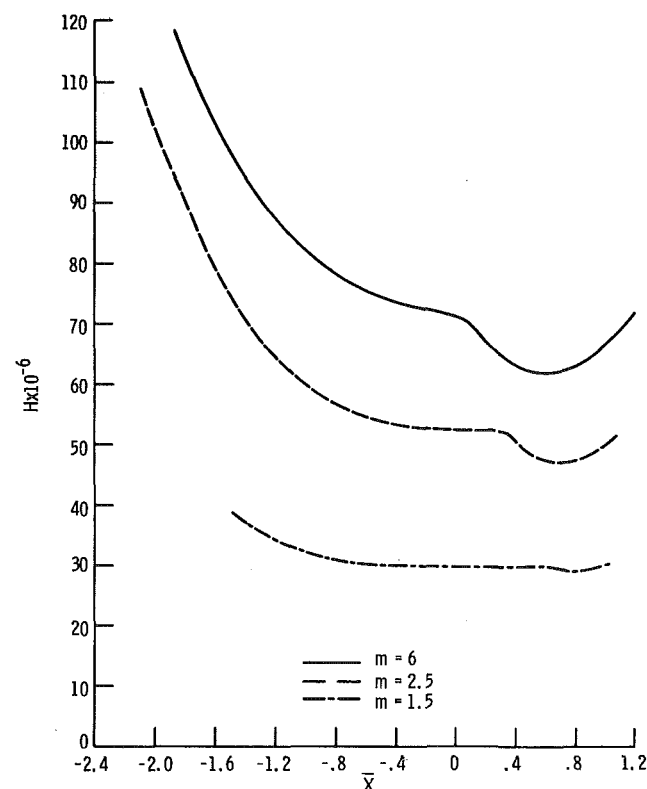


Fig. 8(b) Dimensionless film thickness ($H = h/R_x$) on the $\bar{X}$ axis for three values of dimensionless inlet distance (m). The value of $\bar{Y}$ is held fixed near the axes of symmetry of the contact. The dimensionless speed and load parameters are held fixed as shown in group 3 of Table 2.

lubricant conditions. It is evident from the contour plots that the pressure spike becomes suppressed and the film thickness decreases substantially as the severity of starvation increases.

The results presented in this report, when combined with the findings previously reported [6–8], enable the essential features of starved, and elliptical, elastohydrodynamic conjunctions to be ascertained.

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